

# Telescope Hardware, from the Ground Up — Teaching Notes

CASSA RA 2026-27 · written so it makes sense even on a first read

*How to read this: imagine building a working observatory piece by piece. We start with the tube that catches light, then the mount that aims it, the dome that protects it, the "camera train" that carries the light to the sensor, and the filters that pick which colours we keep. Every part has a plain explanation and one small piece of math.*

## 1. The Optical Tube Assembly (OTA) — the light bucket

**Plain idea:** the OTA is the tube that **collects light and brings it to a focus**. Think of it as a bucket for photons: a bigger opening catches more rain (light).

**Three ways to focus light:** - **Refractor** — uses a **lens** at the front (like a magnifying glass). Simple, sealed, sharp, but big lenses are heavy and give false colour (chromatic aberration). - **Reflector** — uses a **curved mirror** at the back. Cheaper to make big, no colour error. Designs: **Newtonian** (flat secondary sends light to the side), **Cassegrain / Ritchey-Chretien** (secondary sends light back through a hole in the primary; RC is the professional standard, used by Hubble). - **Catadioptric** — mirror **plus** a thin correcting lens (Schmidt-Cassegrain, Maksutov). Compact and popular.

**Inside the tube:** the **primary mirror/lens** (the main collector), the **secondary** (redirects light), **baffles** (black rings that block stray light so the image stays high-contrast), and the **focuser** (moves the camera in/out to get a sharp point).

**The four numbers that define any telescope:** 1. **Aperture D** — the diameter of the main mirror/lens. This is the single most important number. - Light gathered  $\propto D^2$ . *Example:* a 200 mm telescope vs a 100 mm one gathers  $(200/100)^2 = 4\times$  more light  $\rightarrow$  sees fainter stars. 2. **Focal length f** — how far the light travels to focus. Longer **f** = bigger image, narrower view. 3. **Focal ratio  $f/\# = f / D$**  — the "speed." - *Example:*  $D = 200\text{ mm}$ ,  $f = 1000\text{ mm} \rightarrow f/\# = 1000/200 = f/5$ . Small f-number = "fast" = wide field, short exposures (good for faint galaxies). Large f-number = "slow" = high magnification (good for planets). 4. **Resolution (sharpness limit):**  $\theta \approx 1.22 \lambda / D$  (in radians). - *Example:* green light  $\lambda = 550\text{ nm}$ ,  $D = 0.2\text{ m} \rightarrow \theta \approx 1.22 \times 5.5e-7 / 0.2 = 3.4e-6\text{ rad} \approx 0.7\text{ arcsec}$ . Bigger  $D \rightarrow$  finer detail. (On the ground, air blurring called **seeing**,  $\sim 1\text{ arcsec}$ , usually limits you before this.)

## 2. The Mount — how you aim and track

**Plain idea:** the sky appears to rotate (because Earth spins). A mount must **hold the tube steady and slowly turn it** to keep a star centred. If it does not track, stars drift out of view in seconds.

**How fast must it turn?** The sky goes around once per **sidereal day (23 h 56 m)**. - Rate =  $360^\circ / 86164\text{ s} \approx 15.04\text{ arcsec per second}$ . That is the **sidereal rate** every tracking mount matches.

**Two mount types (you have hands-on experience here):** - **Alt-azimuth (Alt-Az)** — moves **up/down (altitude)** and **left/right (azimuth)**, like a camera tripod. Simple and compact. But to follow a star it must drive **both axes at changing speeds**, and the star field slowly **rotates** in the frame (**field rotation**) — long exposures need a "de-rotator." Big professional telescopes are alt-az with computer control. **Dobsonian** is a simple alt-az. - **Equatorial (EQ)** — one axis (the **polar axis**) is tilted to point at the **celestial pole** (near Polaris). Then a **single motor at the sidereal rate** tracks the sky perfectly, with **no field rotation**. Needs careful **polar alignment**. A **German Equatorial Mount (GEM)** uses a counterweight for balance.

**Extras:** **GoTo** (computer database points the scope to any object), **pier/tripod** (rigid support — stability matters more than the tube for sharp images), **guiding** (a small camera makes tiny corrections so long exposures stay pin-sharp).

### 3. The Dome and Enclosure — the "shell"

---

**Plain idea:** the **shell** is the building that protects the telescope from weather and stray light, and opens to the sky when observing.

**Types of enclosure:** - **Classic dome** — a hemisphere with a **slit** that opens; the whole dome **rotates** so the slit follows the telescope. Best wind and light protection. - **Clamshell / roll-off roof** — the roof splits or slides fully away, exposing the whole sky at once. Cheaper, great for wide surveys, less wind shielding.

**Why a shell at all?** - Protects from rain, dust, and sunlight. - **Wind shielding** — wind shake ruins sharp images. - **Thermal control** — the dome is vented/cooled so the air inside matches the night air. Warm air makes wobbling currents ("dome seeing") that blur the image. Professional domes actively manage temperature.

**Two bonus meanings of "shell" you may hear:** - The **OTA shell** = the tube body itself, with internal **baffle rings** and a blackened interior to kill stray light. - **Nested mirror shells** = in **X-ray telescopes** (like Chandra), X-rays only reflect at grazing angles, so mirrors are built as many thin **nested cylindrical shells**, one inside another, to catch more light. Good to know as a contrast to optical mirrors.

---

### 4. The Camera Train (imaging train) — light's path to the sensor

---

**Plain idea:** the **camera train** is the **stack of parts bolted onto the focuser**, in order, that carries and cleans up the light just before it hits the sensor. Order matters, and the spacing (**back-focus**) must be exact or stars look like commas instead of points.

**Typical train, from telescope to camera:** 1. **Focuser** — moves the whole train to reach focus (motorized for precision). 2. **Field flattener / focal reducer** — a lens that makes the focal plane **flat** (so stars at the edge are sharp too) and can shorten the focal length for a wider field. 3. **Filter wheel** — holds several **scientific filters**; rotates to place the chosen filter in the beam (see Section 5). 4. **Off-axis guider (OAG)** — steals a little light to a small **guide camera** that keeps tracking perfect. 5. **Spacers** — precise metal rings to set the exact **back-focus distance** (often ~55 mm for many systems). 6. **Main camera / detector** — the **CCD or CMOS** sensor that records the photons.

**One number that bites everyone — pixel scale:** pixel scale (arcsec/pixel)  $\approx 206 \times (\text{pixel size in } \mu\text{m}) / (\text{focal length in mm})$  - Example: 3.8  $\mu\text{m}$  pixels, focal length 1000 mm  $\rightarrow 206 \times 3.8 / 1000 \approx 0.78$  arcsec/pixel. - Rule of thumb: you want **~2 pixels across the seeing disc** (~1 arcsec) to sample it well. Too big a pixel scale = blocky stars; too small = wasted light and noise.

---

### 5. Scientific Filters — choosing which light to keep

---

**Plain idea:** a filter is a piece of glass that lets **only certain wavelengths pass**. We use them to measure an object's colour, isolate a specific gas, or match a standard system so results are comparable worldwide.

**Two big families:** - **Broadband (photometric) filters** — let through a wide band of colour. Standard systems: - **Johnson-Cousins UBVRI** (ultraviolet, blue, visual, red, infrared). - **SDSS ugriz** and near-IR **JHK**. - *Your link:* these are exactly the bands behind your host-galaxy SED fitting. **SDSS ugriz** (5 bands) and **CSP ugrizYJH** (7 bands) are just measurements through these filters. Each filter gives one point on the galaxy's spectral energy distribution; fitting all the points gives mass, age, dust. -

**Narrowband filters** — let through a very thin slice around one **emission line**, to isolate glowing gas: - **H $\alpha$  (656 nm)** — hydrogen, traces star formation and HII regions. - **OIII (501 nm)** — doubly ionized oxygen, planetary nebulae and shocks. - **SII (672 nm)** — sulfur, supernova remnants.

**How they physically work:** most scientific filters are **interference (dichroic) filters** — many thin coated layers that let the wanted wavelength through and **reflect** the rest (not absorb). This makes them sharp-edged and efficient.

**Why filters are calibrated so carefully:** to turn "counts on my chip" into a real magnitude you need the **exact filter response**, plus **standard stars**, a **zero point**, and **color terms** (small corrections

when your filter is not perfectly identical to the standard one). Same photometry chain from the Observatory notes, now you see where the filter fits in.

## Putting it together (say this as a story in the viva)

Light from a galaxy enters the **OTA**, where the **aperture** sets how faint I can go and the **f-ratio** sets my field. The **mount** tracks it at the **sidereal rate** so it stays fixed on the sensor. The **dome** keeps wind and warm air from blurring the image. In the **camera train**, a **field flattener** sharpens the edges, a **filter** in the wheel selects the band (say SDSS *r*), and the **CCD/CMOS** records it. I calibrate with **bias/dark/flat** and **standard stars**, get a magnitude in each filter, and those magnitudes become the **SED** I fit for host-galaxy age. That last step is my research.

## Quick-recall (flash-card)

- **OTA:** refractor (lens) / reflector (mirror: Newtonian, Cassegrain, RC) / catadioptric (SCT, Mak). Aperture  $D$ , focal length  $f$ ,  $f/\# = f/D$ , resolution  $\theta \approx 1.22\lambda/D$ , light  $\propto D^2$ .
- Example:  $D=200\text{mm}$ ,  $f=1000\text{mm} \rightarrow f/5$ ;  $\theta \approx 0.7''$  at  $550\text{nm}$ .
- **Mount:** sidereal rate  $\approx 15.04 \text{ arcsec/s}$ . Alt-Az (2 axes, field rotation) vs Equatorial (1 polar axis, no rotation, needs polar align). GoTo, guiding, rigid pier.
- **Shell/dome:** rotating slit dome vs roll-off/clamshell; protects + shields wind + thermal control ("dome seeing"). OTA shell = baffled tube. X-ray = nested grazing-incidence shells.
- **Camera train:** focuser  $\rightarrow$  flattener/reducer  $\rightarrow$  filter wheel  $\rightarrow$  off-axis guider  $\rightarrow$  spacers (back-focus)  $\rightarrow$  camera. Pixel scale  $\approx 206 \times \text{pixel}/\mu\text{m} / f_{\text{mm}}$ ; aim  $\sim 2 \text{ px per arcsec}$ .
- **Filters:** broadband UBVRI / ugriz / JHK (photometry  $\rightarrow$  SED); narrowband H $\alpha$ /OIII/SII (emission gas). Interference/dichroic (reflect, not absorb). Need zero point + standards + color terms.

## Likely viva Qs

1. What do aperture and focal ratio each control?
2. Why must a mount track, and at what rate? Alt-Az vs equatorial difference?
3. Why do observatories bother with a rotating dome and thermal control?
4. Walk me through a camera/imaging train from focuser to sensor.
5. Difference between broadband and narrowband filters? How do filters connect to your SED fitting?